Samridhi Vibhanshu

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Education

Vellore Institute of Technology, Vellore Aug. 2023 – Present
B.Tech. in Computer Science and Engineering, Specialization in Business Studies *CGPA: 9.81/10*

Delhi Public School, Ranchi *Class XII (CBSE): 96.40%* Apr. 2023

Projects

Agentic Bug Repair | *Java 17, JDK Compiler API, Linux namespaces, JavaScript* GitHub | 2026
- Built an agent that repairs failing Java builds: indexes the repo into a **code graph**, ranks suspect methods from call-graph coverage, stack traces and hybrid **BM25/vector retrieval**, and tests candidate patches in parallel.
- Each patch runs in a **sandbox** (no network, resource limits, watchdog); a pull request is opened only if every test passes on a clean checkout. HMAC-verified GitHub webhook, bounded job queue, live dashboard; **148 tests**.

Order Service | *Java 17, java.util.concurrent, JDK HTTP server* GitHub | 2026
- E-commerce order backend over a JSON REST API that **never oversells**: per-SKU locks taken in sorted order prevent deadlock; 64 threads racing for the last 50 units always yield exactly 50 orders.
- **Idempotency keys** stop double charges on retry; state machine and outbox log; **30 tests** incl. 9,600 random workflows.

Mini-SQS and Route Planner | *Java 17, no dependencies* GitHub | GitHub (routes) | 2026
- **SQS-style queue**: visibility timeouts, DLQs, FIFO dedup, long polling; **26 tests** incl. 600 randomized sequences.
- Routing with Dijkstra, **A***, Held–Karp and 2-opt, checked against Floyd–Warshall and brute force (**19 tests**).

IS Trust Decay Prediction | *Python, Flask, NetworkX, Scikit-learn, SQLite, pytest* GitHub | 2026
- Audit events feed a **temporal knowledge graph** whose node, history and graph features drive a **Random Forest** that forecasts each asset's trust decay 5 days ahead; chronological, purged splits prevent look-ahead leakage.
- Daily top-5 audit list caught **47%** high-risk assets vs. **31%** at random (simulated); Flask **REST API**; **68 tests**.

Risk-Constrained Emergency Control | *Python, Scikit-learn* GitHub | 2026
- Ventilation control that tries a small **reversible test action** first and escalates only once it is confirmed safe.
- Safety-gated classifier on 5,000 simulated records: **0.8% false-safe rate**, **92.9% harmful-class recall**.

Hybrid Real-Time Fall Detection | *Python, Scikit-learn, C++, ESP32, MPU6050, Blynk IoT* 2026
- Random Forest on SisFall (3 s windows, 50% overlap); **Edge AI wearable** on ESP32 + MPU6050 that confirms falls by combining predictions with acceleration and orientation thresholds and alerts caregivers via Blynk IoT.

Patent Applications (6 through the VIT IPR Cell; 5 published)

Autonomous Wildfire Prediction and Response System *App. No. 202641083188*	Published, Jul. 2026
Distributed Tremor Detection and Hazard Response in Mining *App. No. 202641075003*	Published, Jun. 2026
Confidence-Weighted Opinion Lineage for Sentiment Aggregation *App. No. 202641115708*	Published, Sep. 2026
Predictive Trust Assessment and Audit Prioritization of IS Assets *App. No. 202641108535*	Published, 2026
Risk-Constrained Emergency Control via Reversible Micro-Interventions	Published, 2026
AI-Enabled Predictive Railway Collision Avoidance	Filed, 2026

Papers (submitted to IEEE Access, 2026; under review)

Physics-Shielded Conformal Learning for Network-Aware Railway Collision-Risk Assessment Under Degraded Observations
Forecasting Information-System Trust Decay on Temporal Knowledge Graphs With Delayed-Feedback Conformal Audit Selection
Confidence-Weighted Opinion Lineage for Organic Sentiment Estimation Under Synthetic and Coordinated Amplification

Leadership

Computer Society of India (CSI-VIT) Apr. 2024 – Present
Board Member (Core Lead) | Full Stack Developer *Vellore, Tamil Nadu*
- Promoted to **Board Member (Core Lead)** in Sep. 2025; lead development, design, operations and outreach teams.
- Contributed to the Laser Tag platform (React.js, Tailwind CSS); built a GitHub contribution visualization.

Programme Representative, VIT *Student–faculty liaison for 200+ students* Jul. 2024 – Present

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, SQL, C, C++, R, HTML, CSS
Frameworks/Libraries: React.js, Next.js, Redux, Tailwind CSS, Flask, Scikit-learn, Pandas, NumPy, NetworkX, pytest
Tools/Platforms: Git, GitHub Actions, Linux, Postman, SQLite, OCI, ESP32, MPU6050, Blynk IoT
Concepts: DSA, OOP, Concurrency, DBMS, REST APIs, Machine Learning, RAG, Edge AI, IoT
Certifications: OCI 2025 Certified Generative AI Professional; OCI 2025 Certified AI Foundations Associate